Comparative Sequence and Structure Analysis of APP, PSEN1 and PSEN2: Conservation of Alzheimer's Disease Residues and Drug-Binding Sites Across Laboratory Model Organisms

A reproducible dry-lab bioinformatics study

Data retrieved 29 October to 3 November 2026. Report generated 03 November 2026.

Interactive results: https://nozheimers.vercel.app and https://nozheimers.vercel.app/explorer.html

Code and data: https://github.com/Kukyos/Nozheimers

Abstract

Background: mutations in APP, PSEN1 and PSEN2 cause early-onset Alzheimer's disease, and drug candidates are tested in animals before humans. Aim: to determine which laboratory organisms carry proteins close enough to the human ones for disease-mutation and drug studies. Methods: with the human proteins as the reference, we aligned their orthologs from four mammals (multiple alignment, Clustal Omega) and from 12 laboratory organisms (pairwise), superposed AlphaFold structures, mapped curated ClinVar pathogenic variants, and measured conservation of the residues that contact nine drugs in experimental PDB structures. Results: identity to human in mouse, rat, dog and cattle was 96.6-97.4% for APP, 91.7-94.0% for PSEN1 and 95.3-97.5% for PSEN2. 111 of 112 mapped pathogenic variants lie at residues identical in all five mammals. Mouse PSEN1 keeps 68 of 69 disease residues and all 5 drug pockets; mouse APP keeps all 9 disease residues but differs at 3 of 42 amyloid-beta residues. Conclusion: non-human primates, pig, cattle and dog carry essentially human versions of all three proteins; rodents suit presenilin and gamma-secretase drug studies, but amyloid-beta studies in rodents need the human APP sequence.

1. Introduction

Alzheimer's disease is the most common cause of dementia. A small fraction of cases are inherited and begin early, caused by mutations in APP [1], PSEN1 [2] or PSEN2 [3, 4]. APP is cut twice to release amyloid-beta, the peptide that accumulates in plaques [5]. Presenilin forms the catalytic core of gamma-secretase, which makes the second cut with two aspartate residues [6, 7].

Drugs and disease mechanisms are studied in animals before humans. Such a study transfers to patients only if the animal's protein has the same residues where it matters: at the sites mutated in patients, in the functional core, and where a drug binds. Sequence comparison counts the differences; structure comparison shows where they sit.

Research question: how conserved are APP, PSEN1 and PSEN2 between humans and laboratory organisms, and which organisms keep the human disease residues, functional core and drug-binding residues well enough to serve as test models?

2. Input

The human protein is the reference in every comparison; the other organisms are the inputs tested against it.

Table 1. Human reference proteins (MANE Select). The UniProt canonical sequences are identical to these RefSeq proteins, so residue numbers agree across databases.

Gene	Protein	Length (aa)	RefSeq protein	Transcript	UniProt	NCBI Gene
APP	amyloid-beta precursor protein isoform a precursor	770	NP_000475.1	NM_000484.4	P05067	351
PSEN1	presenilin-1 isoform I-467	467	NP_000012.1	NM_000021.4	P49768	5663
PSEN2	presenilin-2 isoform 1	448	NP_000438.2	NM_000447.3	P49810	5664

Table 2. Laboratory organisms compared with the human reference.

Group	Organism	Species	NCBI Taxon ID
Primates	Chimpanzee	Pan troglodytes	9598
Primates	Rhesus macaque	Macaca mulatta	9544
Rodents	Mouse	Mus musculus	10090
Rodents	Rat	Rattus norvegicus	10116
Other mammals	Dog	Canis lupus familiaris	9615
Other mammals	Pig	Sus scrofa	9823
Other mammals	Cattle	Bos taurus	9913
Birds	Chicken	Gallus gallus	9031
Amphibians	Western clawed frog	Xenopus tropicalis	8364
Fish	Zebrafish	Danio rerio	7955
Invertebrates	Fruit fly	Drosophila melanogaster	7227
Invertebrates	C. elegans	Caenorhabditis elegans	6239

The online explorer extends the comparison to any species with an NCBI ortholog call (832-866 species, depending on the gene).

Databases and tools: NCBI Gene, RefSeq and Datasets [8, 9, 10]; NCBI BLAST [11]; Clustal Omega via the EMBL-EBI Job Dispatcher [12, 13]; ClinVar [14]; UniProt [15]; AlphaFold DB [16, 17]; RCSB Protein Data Bank [18]; pSSdb, the Protein Secondary Structure Database (https://pssdb.mimuw.edu.pl); Biopython [19]; 3Dmol.js [20]; matplotlib [21].

3. Data and methods

All rules were fixed before results were seen. Every decision is logged in 01_metadata/methods_log.md, and every download is cached with its access date.

Figure 1. Analysis workflow (each step writes to its numbered project folder)

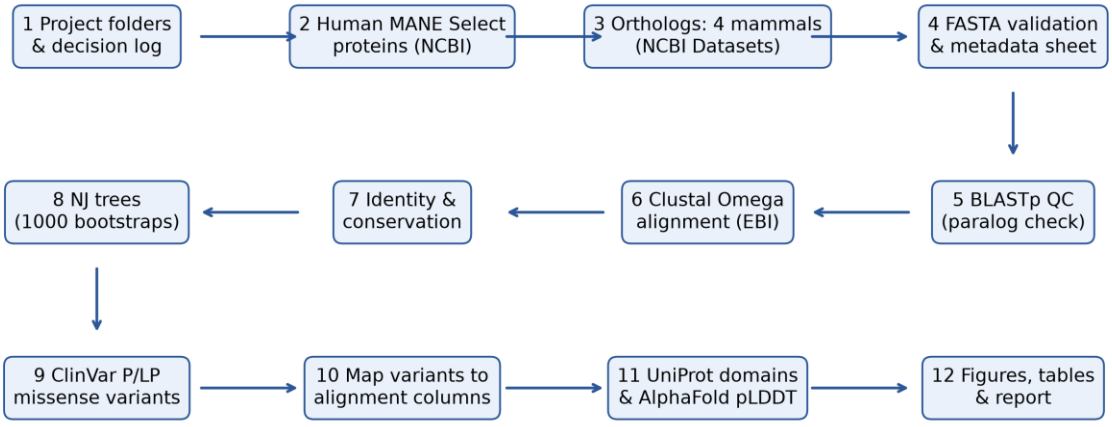


Figure 1. Workflow of the five-mammal analysis (pipeline.py), from sequence retrieval to interpretation.

3.1 Five-mammal comparison

- Human reference: the MANE Select transcript and protein from the NCBI Gene product report.
- Orthologs in mouse, rat, dog and cattle: NCBI ortholog calls. Among 126 candidate RefSeq proteins, the isoform with the highest global alignment score to human was kept (BLOSUM62 [22], gap open -11, extend -1), preferring a curated NP_ record within 5% of the best score.
- Quality control: FASTA validation, and BLASTp of each human protein against RefSeq protein to confirm that the top hit per species is the ortholog, not a paralog.
- Multiple alignment: Clustal Omega, protein, default parameters, one alignment per gene.
- Identity = identical residues / columns where both sequences have a residue. Coverage = those columns / human length.
- Trees: neighbour-joining [23] on p-distance with 1000 bootstrap replicates [24].
- ClinVar: missense records classified Pathogenic or Likely pathogenic, review status "criteria provided" or better, with an Alzheimer condition listed; mapped only when the transcript and reference residue match the human reference.

3.2 Laboratory organisms and structures

- Orthologs for 12 organisms from NCBI ortholog calls, mapped to UniProt. All UniProt entries linked to the gene were aligned to human and the best full-length match was kept. NCBI makes no call for fruit fly and C. elegans, so reviewed UniProt homologs were used (Appl and apl-1 for APP; Psn and sel-12, each a co-ortholog of both PSEN1 and PSEN2).
- Global pairwise alignment to human (BLOSUM62, gap open -11, extend -1).
- Structure comparison: the AlphaFold DB models of the human and organism proteins were superposed on C-alpha atoms (Kabsch [25]) using residues with pLDDT of at least 70 in both models. APP, a multi-domain protein with flexible linkers, was fitted on its best-fitting domain (E2, residues 374-565).

- Functional core: amyloid-beta (APP 672-713) for APP; the UniProt membrane-spanning helices for PSEN1 and PSEN2, with the catalytic aspartates PSEN1 D257/D385 and PSEN2 D263/D366.
- Drug pockets: human residues with any heavy atom within 4.5 angstrom of the drug in experimental PDB complexes (Table 9).
- Secondary structure: helix, sheet and coil assigned from each 3D model with a DSSP-style hydrogen-bond method [26], the same approach pSSdb applies to AlphaFold models, so pSSdb entries for the same UniProt accessions serve as an independent cross-check.
- Verdict, stated in advance so it can be challenged. Strong model: at least 90% identity, at least 95% of disease residues kept, at least 95% of the core identical, catalytic residues kept. Usable with care: at least 70% identity, catalytic residues kept, at least 85% of disease residues kept. Otherwise: distant model.
- Checks: human against itself gives 100% identity and 0 angstrom RMSD; the browser implementation reproduces the Python results exactly (site/test_live.js); MRK-560, reported as PSEN1-selective, has the pocket that differs most between human PSEN1 and PSEN2 (T281P, L282I).

Table 3. Tools, versions and parameters (01_metadata/analysis_settings.csv).

step	tool_or_resource	version_or_endpoint	parameters	date
2 Human references	NCBI Datasets v2 gene product_report; E-utilities efetch	https://api.ncbi.nlm.nih.gov/datasets/v2	MANE Select transcript and protein per gene	2026-11-03
3 Orthologs	NCBI Datasets v2 gene orthologs + product_report; Biopython PairwiseAligner	https://api.ncbi.nlm.nih.gov/datasets/v2/gene/id/{id}/orthologs	global alignment, BLOSUM62, gap open -11, extend -1; NP_ preferred within 5% of best score	2026-11-03
5 BLAST QC	NCBI BLAST URL API, blastp	BLASTP 2.17.0+	db refseq_protein; Entrez query "txid10090[ORGN] OR txid10116[ORGN] OR txid9615[ORGN] OR txid9913[ORGN]"; hitlist 100; matrix BLOSUM62; gap open/extend 11/1; expect 10	2026-11-03
6 Multiple alignment	EMBL-EBI Job Dispatcher Clustal Omega REST	CLUSTAL O(1.2.4)	stype=protein, outfmt=clustal_num, order=input, all other parameters default	2026-11-03
7 Identity/conservation	pipeline.py (Python)		identity = identical / columns where both sequences have a residue; coverage = those columns / human length	2026-11-03
8 Phylogeny	Biopython DistanceTreeConstructor (neighbour-joining)	Biopython 1.88	p-distance (gap columns excluded pairwise); 1000 bootstrap replicates, seed 1; midpoint-rooted for display; no substitution-model selection	2026-11-03
9 ClinVar	NCBI E-utilities esearch/esummary, db=clinvar	https://eutils.ncbi.nlm.nih.gov/entrez/eutils/	term "<GENE>[gene] AND missense_variant[molecular consequence]";	2026-11-03

			selection rule in methods_log.md	
11 Domains/structure	UniProt REST; AlphaFold DB API	rest.uniprot.org; alphafold.ebi.ac.uk/api	reviewed human entry; features transferred only if sequence identical; pLDDT from CA B-factor	2026-11-03
12 Figures	matplotlib	3.11.1	300 dpi PNG	2026-11-03

4. Output (results)

4.1 Sequence retrieval and quality control

All 15 selected proteins passed validation. BLASTp confirmed the APP ortholog as the top hit in every species. The searches for PSEN1 and PSEN2 were still queued at NCBI when this report was generated and are marked pending; re-running pipeline.py fills them in.

4.2 Conservation across five mammals

Table 4. Pairwise identity of each species to the human reference (from the Clustal Omega alignment).

Gene	Species	Accession	Identity %	Coverage %	Shared columns	Gap columns
APP	Mus musculus	NP_001185752.1	96.62	100.0	770	0
APP	Rattus norvegicus	NP_062161.1	96.88	100.0	770	0
APP	Canis lupus familiaris	NP_001006601.2	96.88	100.0	770	0
APP	Bos taurus	XP_005201194.1	97.4	100.0	770	0
PSEN1	Mus musculus	NP_001349200.1	92.87	99.14	463	8
PSEN1	Rattus norvegicus	NP_062036.2	92.66	99.14	463	9
PSEN1	Canis lupus familiaris	NP_001002943.2	93.99	99.79	466	1
PSEN1	Bos taurus	NP_777146.1	91.65	100.0	467	11
PSEN2	Mus musculus	NP_001122077.1	95.98	100.0	448	0
PSEN2	Rattus norvegicus	NP_112349.2	95.31	100.0	448	0
PSEN2	Canis lupus familiaris	NP_001271394.1	95.76	100.0	448	0
PSEN2	Bos taurus	NP_776865.2	97.54	100.0	448	1

APP: 94.3% of human residues are identical in all five species; PSEN1: 85.4% of human residues are identical in all five species; PSEN2: 91.1% of human residues are identical in all five species.

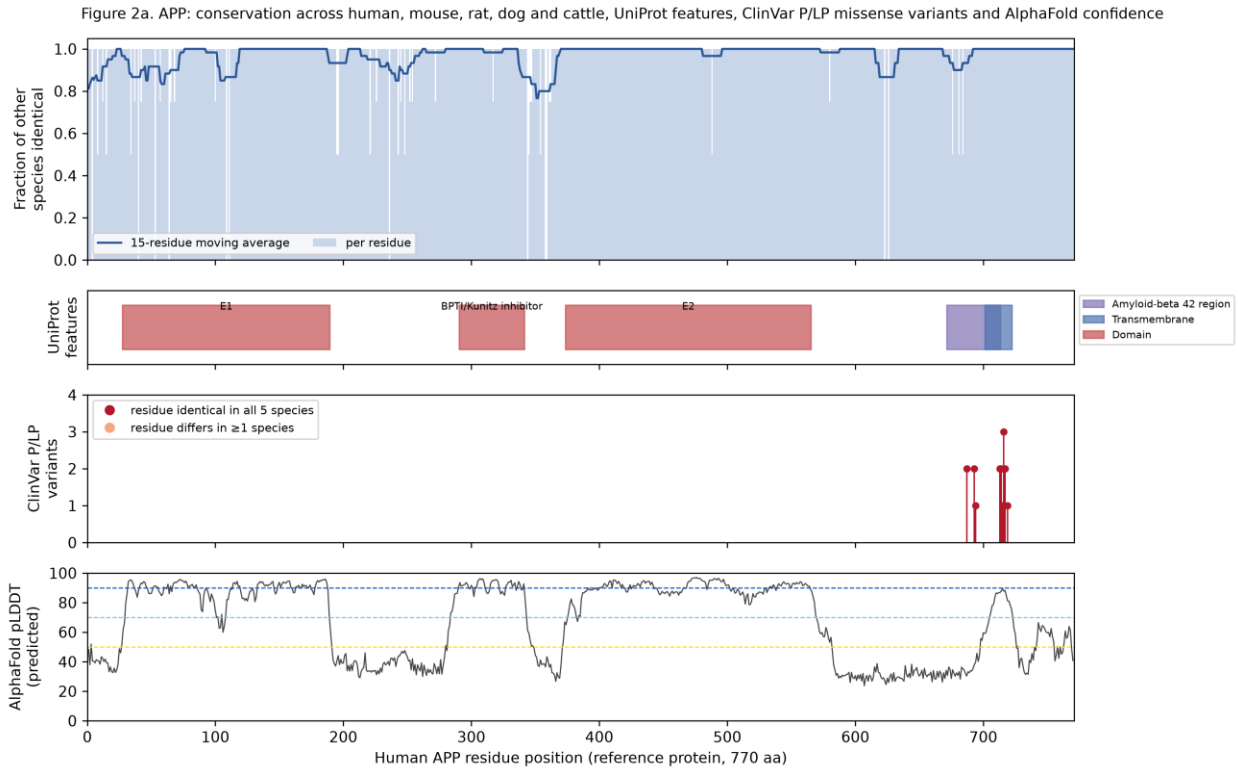


Figure 2a. APP: per-residue conservation across five species, UniProt features and ClinVar pathogenic variants.

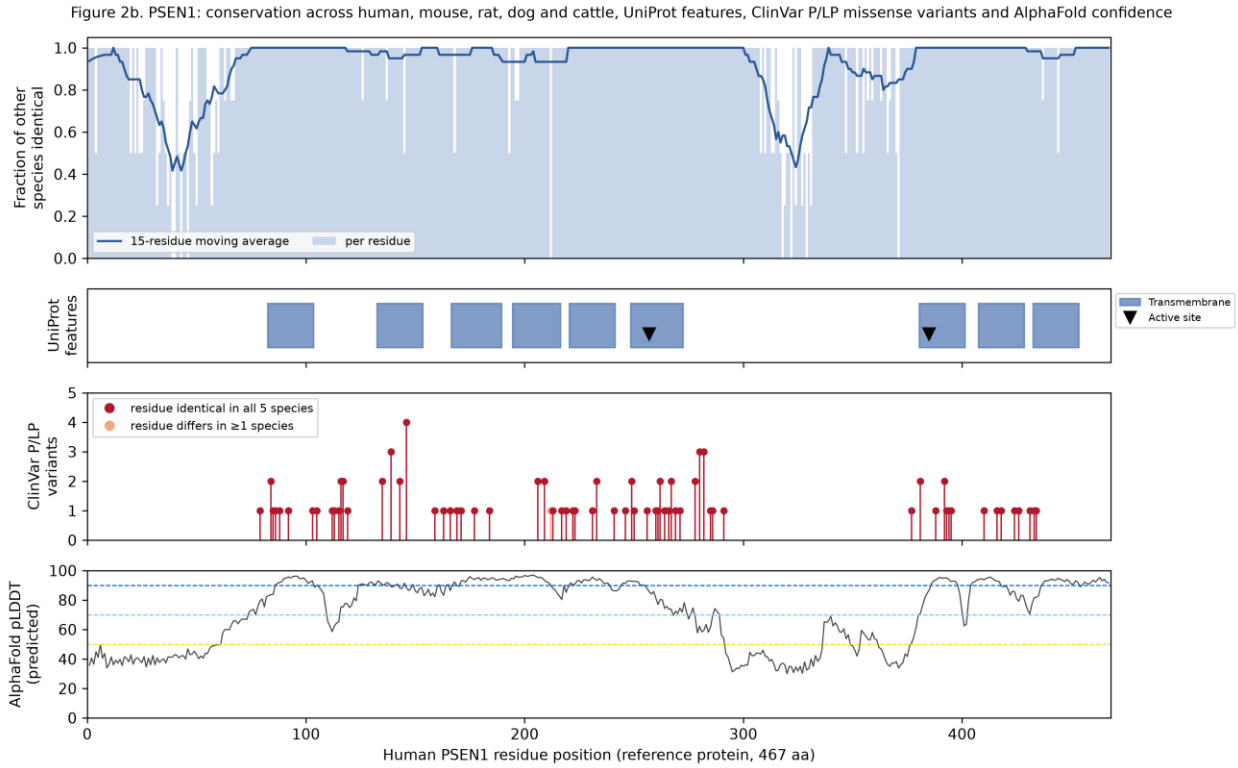


Figure 2b. PSEN1: per-residue conservation across five species, UniProt features and ClinVar pathogenic variants.

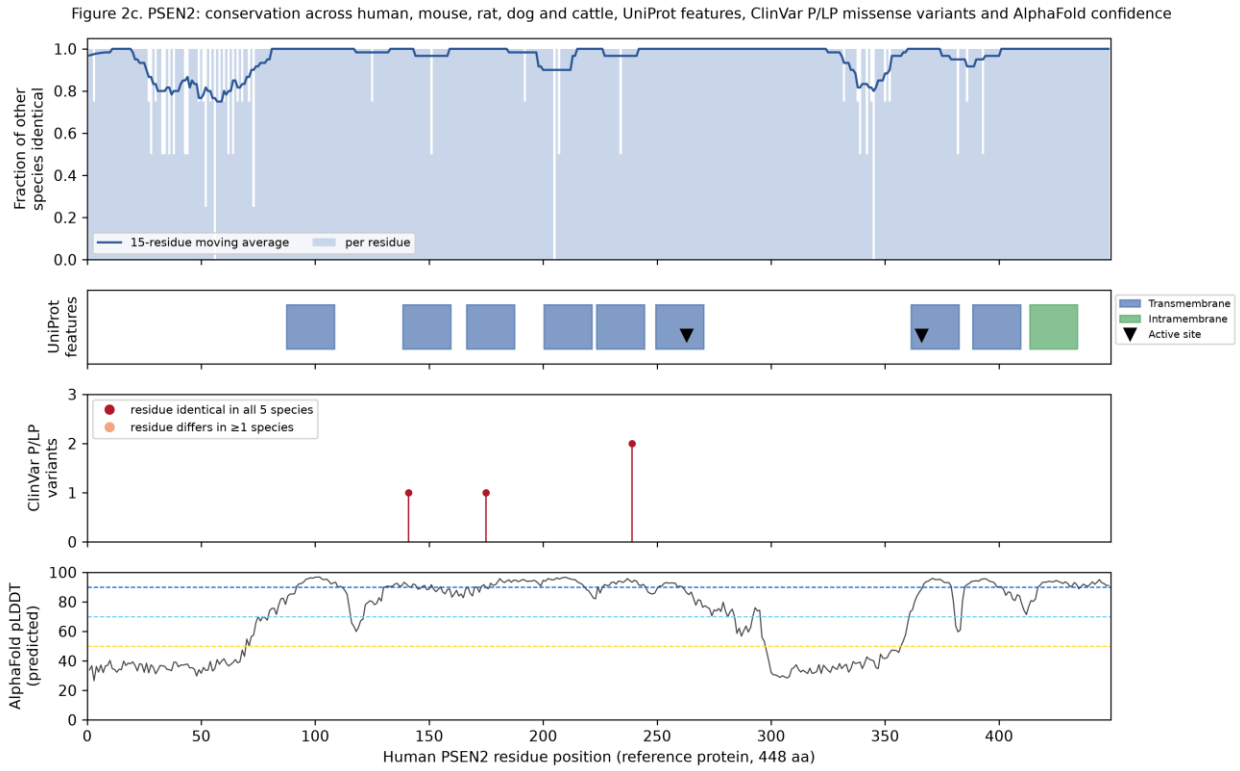


Figure 2c. PSEN2: per-residue conservation across five species, UniProt features and ClinVar pathogenic variants.

4.3 Disease variants

ClinVar missense records: APP 276, PSEN1 361, PSEN2 170. Of these, APP 16, PSEN1 92, PSEN2 4 met the selection rule, and all were mapped. 111 of 112 lie at residues identical in all five mammals.

Table 5. Curated pathogenic variants and conservation. The full list with ClinVar links is in 07_variants/variant_mapping.csv.

Gene	Variants	Residues	Variants at fully conserved residues	all residues fully conserved %
APP	16	9	16	94.3
PSEN1	92	69	91	85.4
PSEN2	4	3	4	91.1

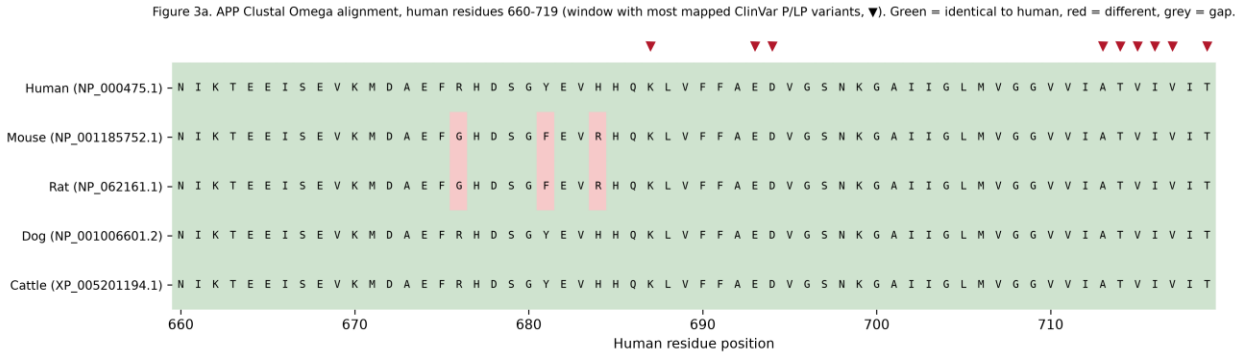


Figure 3a. APP: alignment window around the densest cluster of pathogenic variants.

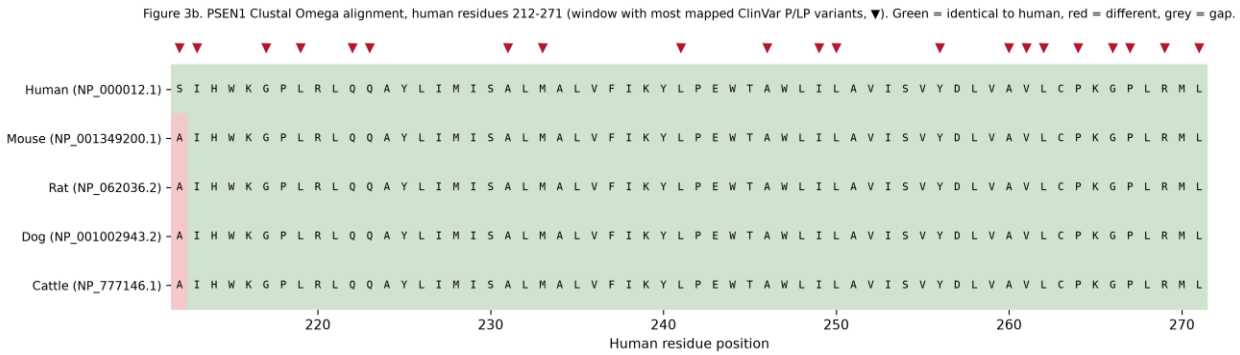


Figure 3b. PSEN1: alignment window around the densest cluster of pathogenic variants.

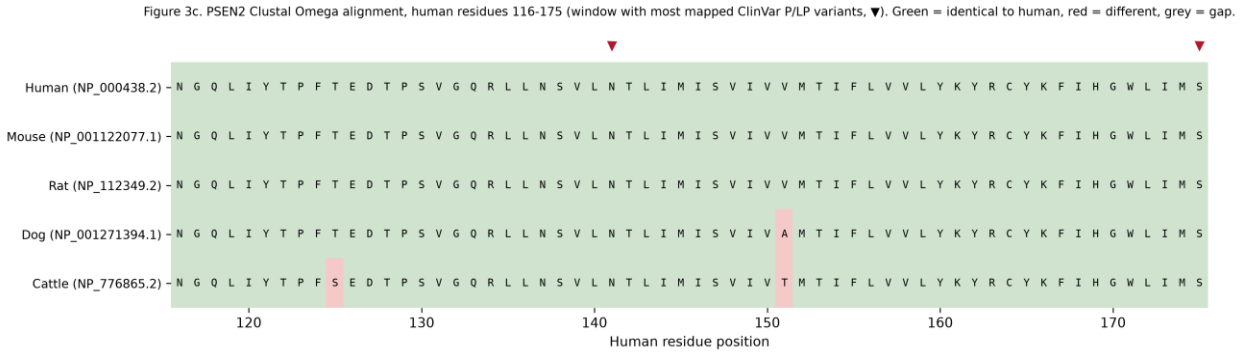


Figure 3c. PSEN2: alignment window around the densest cluster of pathogenic variants.

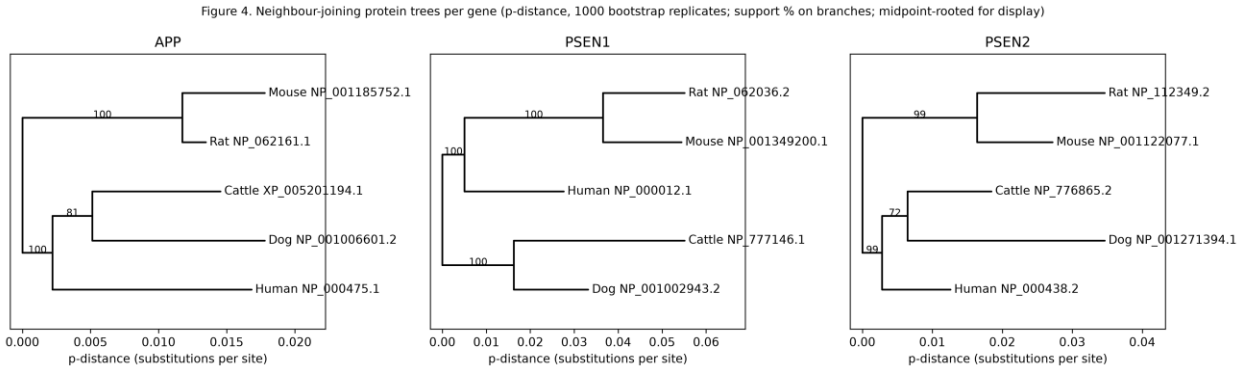


Figure 4. Neighbour-joining gene trees (p-distance, 1000 bootstraps). These are gene trees, not disease trees.

4.4 3D structure comparison

In the online explorer the organism model is superposed on the human model and can be rotated, coloured by difference, confidence, secondary structure, disease residues or drug pocket, and queried residue by residue.



Figure 5a. Human APP (blue) and mouse App (amber) AlphaFold models superposed on the E2 domain; magenta marks residues that differ. Screenshot of the model-organism explorer.

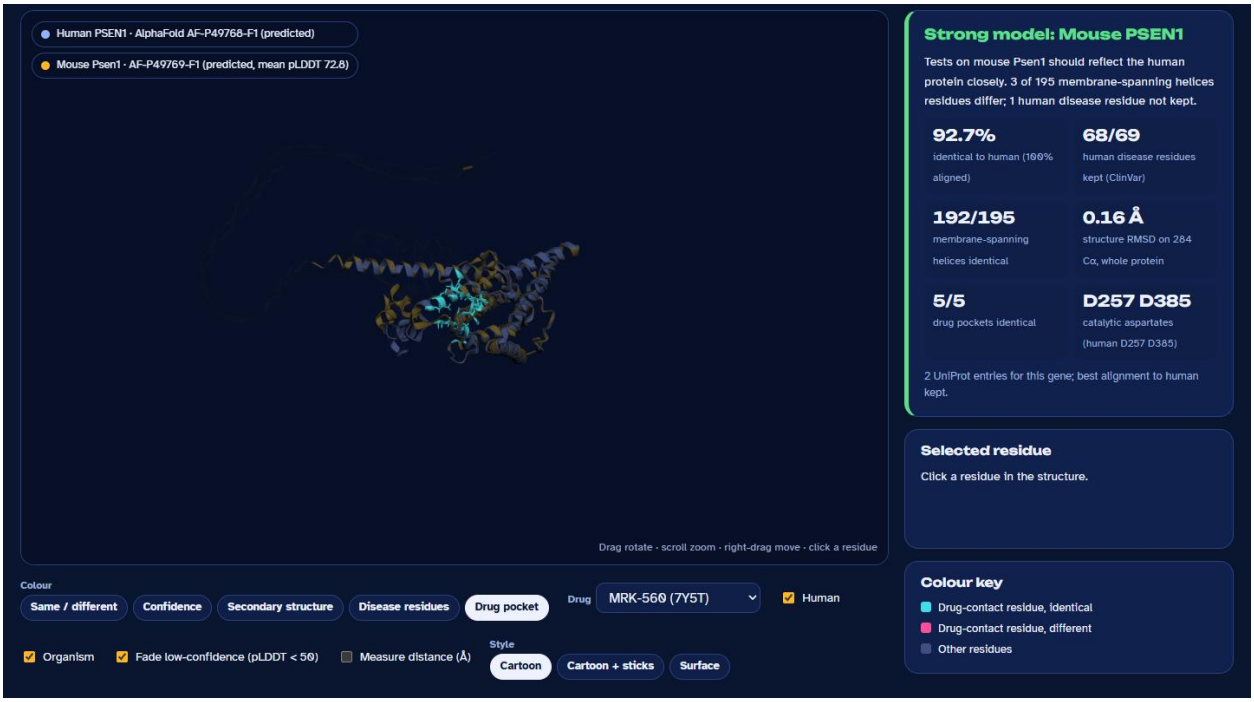


Figure 5b. Human and mouse PSEN1 superposed (0.16 angstrom RMSD); cyan sticks mark the residues that contact MRK-560 in PDB 7Y5T, all identical in mouse. Screenshot of the model-organism explorer.

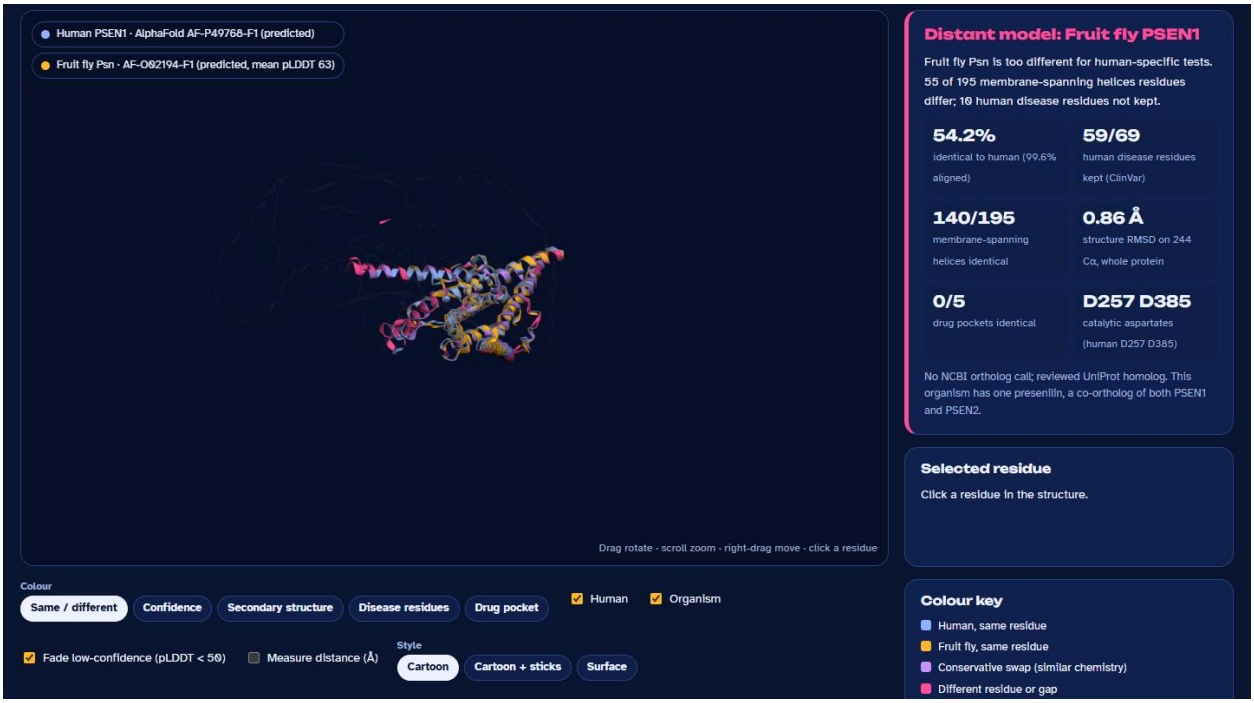


Figure 5c. Human PSEN1 and fruit fly Psn: the membrane core superposes closely (0.86 angstrom) despite 54% identity, but disease and drug-contact residues differ. Screenshot of the model-organism explorer.

4.5 Laboratory organisms

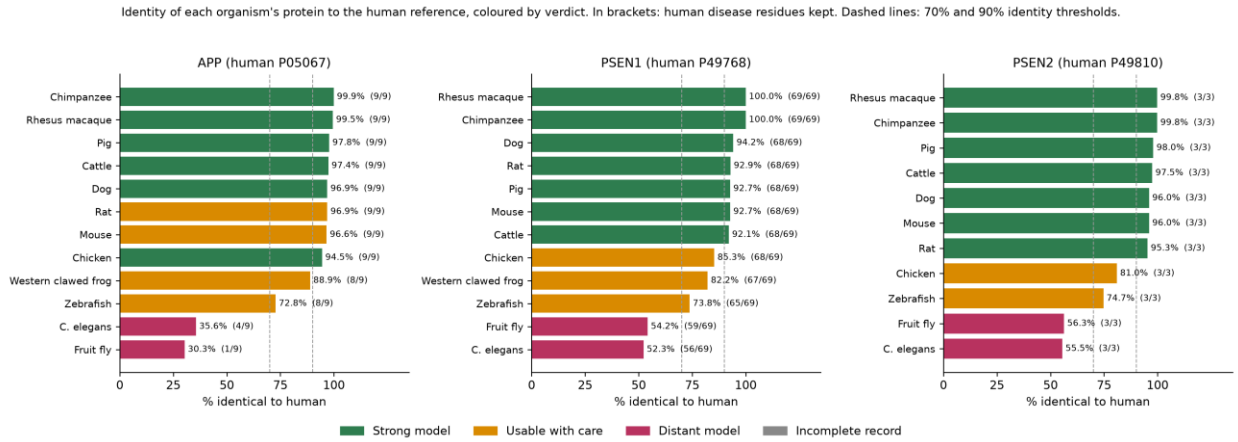


Figure 6. Identity of each laboratory organism's protein to the human reference, coloured by verdict.

Table 6. Human APP (P05067) against laboratory organisms. Core = amyloid-beta (APP 672-713). RMSD in angstrom over the number of C-alpha atoms in brackets.

Organism	UniProt	Identity %	Disease residues kept	Core identical	RMSD (CA)	Drug pockets identical	Verdict
Chimpanzee	Q5IS80	99.9	9/9	42/42	1.0 (188)	4/4	Strong model
Rhesus macaque	G7MNB1	99.5	9/9	42/42	0.43 (187)	4/4	Strong model
Pig	P79307	97.8	9/9	42/42	0.84 (180)	4/4	Strong model
Cattle	A0A3Q1LUG2	97.4	9/9	42/42	0.69 (188)	4/4	Strong model

Rat	P08592	96.9	9/9	39/42	0.83 (184)	1/4	Usable with care (needs human APP for amyloid-beta work)
Dog	A0A8I3P9H6	96.9	9/9	42/42	0.62 (186)	4/4	Strong model
Mouse	P12023	96.6	9/9	39/42	0.14 (189)	1/4	Usable with care (needs human APP for amyloid-beta work)
Chicken	Q9DGJ7	94.5	9/9	42/42	0.59 (187)	4/4	Strong model
Western clawed frog	Q6DJB6	88.9	8/9	37/42	0.56 (180)	0/4	Usable with care (needs human APP for amyloid-beta work)
Zebrafish	Q6NUZ1	72.8	8/9	34/42	1.11 (188)	1/4	Usable with care (needs human APP for amyloid-beta work)
C. elegans	Q10651	35.6	4/9	8/42	1.68 (145)	0/4	Distant model
Fruit fly	P14599	30.3	1/9	12/42	2.98 (186)	0/4	Distant model

Table 7. Human PSEN1 (P49768) against laboratory organisms. Core = membrane-spanning helices. RMSD in angstrom over the number of C-alpha atoms in brackets.

Organism	UniProt	Identity %	Disease residues kept	Core identical	RMSD (CA)	Drug pockets identical	Verdict
Chimpanzee	H2Q8K1	100.0	69/69	195/195	0.26 (273)	5/5	Strong model
Rhesus macaque	A0A5K1VUH6	100.0	69/69	195/195	0.25 (271)	5/5	Strong model
Dog	A0A8I3RZ86	94.2	68/69	194/195	0.23 (270)	5/5	Strong model
Rat	P97887	92.9	68/69	192/195	0.17 (282)	5/5	Strong model
Mouse	P49769	92.7	68/69	192/195	0.16 (284)	5/5	Strong model
Pig	Q0MS44	92.7	68/69	192/195	0.23 (272)	5/5	Strong model
Cattle	A7MBA9	92.1	68/69	188/195	0.23 (265)	5/5	Strong model
Chicken	Q4JIM4	85.3	68/69	190/195	0.18 (282)	5/5	Usable with care
Western clawed frog	A0A803KHG8	82.2	67/69	186/195	0.23 (264)	5/5	Usable with care
Zebrafish	F8W381	73.8	65/69	176/195	0.28 (262)	2/5	Usable with care
Fruit fly	O02194	54.2	59/69	140/195	0.86 (244)	0/5	Distant model
C. elegans	P52166	52.3	56/69	128/195	0.71 (273)	0/5	Distant model

Table 8. Human PSEN2 (P49810) against laboratory organisms. Core = membrane-spanning helices. RMSD in angstrom over the number of C-alpha atoms in brackets.

Organism	UniProt	Identity %	Disease residues kept	Core identical	RMSD (CA)	Drug pockets identical	Verdict
Chimpanzee	A0A2I3SVT3	99.8	3/3	168/168	0.29 (273)	-	Strong model
Rhesus macaque	F7A2D1	99.8	3/3	168/168	0.24 (271)	-	Strong model
Pig	Q0MS45	98.0	3/3	166/168	0.09 (285)	-	Strong model
Cattle	Q9XT96	97.5	3/3	166/168	0.11 (284)	-	Strong model
Mouse	Q3U4P5	96.0	3/3	163/168	0.3 (274)	-	Strong model

Dog	A0A8I3NJ39	96.0	3/3	166/168	0.22 (270)	-	Strong model
Rat	O88777	95.3	3/3	163/168	0.15 (283)	-	Strong model
Chicken	A0A8V0Z5X8	81.0	3/3	155/168	0.29 (272)	-	Usable with care
Zebrafish	Q90ZE4	74.7	3/3	145/168	0.18 (276)	-	Usable with care
Fruit fly	O02194	56.3	3/3	123/168	0.6 (244)	-	Distant model
C. elegans	P52166	55.5	3/3	109/168	0.46 (273)	-	Distant model

No ortholog listed for: Western clawed frog.

4.6 Drug-binding pockets

Table 9. Drugs in experimental human structures and the organisms whose drug-contact residues are identical to human. This is pocket conservation, not a docking simulation.

Drug	Type	Target	PDB	Method, resolution	Pocket residues	Organisms with identical pocket
Avagacestat	gamma-secretase inhibitor	PSEN1	6LQG	Electron Microscopy, 3.1 A	19	Chimpanzee, Rhesus macaque, Mouse, Rat, Dog, Pig, Cattle, Chicken, Western clawed frog
Semagacestat	gamma-secretase inhibitor	PSEN1	6LR4	Electron Microscopy, 3.0 A	16	Chimpanzee, Rhesus macaque, Mouse, Rat, Dog, Pig, Cattle, Chicken, Western clawed frog
L-685,458	transition-state inhibitor	PSEN1	7C9I	Electron Microscopy, 3.1 A	24	Chimpanzee, Rhesus macaque, Mouse, Rat, Dog, Pig, Cattle, Chicken, Western clawed frog, Zebrafish
E2012	gamma-secretase modulator	PSEN1	7D8X	Electron Microscopy, 2.6 A	9	Chimpanzee, Rhesus macaque, Mouse, Rat, Dog, Pig, Cattle, Chicken, Western clawed frog, Zebrafish
MRK-560	PSEN1-selective inhibitor	PSEN1	7Y5T	Electron Microscopy, 2.9 A	22	Chimpanzee, Rhesus macaque, Mouse, Rat, Dog, Pig, Cattle, Chicken, Western clawed frog
Aducanumab	anti-amyloid antibody	APP	6CO3	X-Ray Diffraction, 2.384 A	6	Chimpanzee, Rhesus macaque, Dog, Pig, Cattle,

						Chicken
Solanezumab	anti-amyloid antibody	APP	4XXD	X-Ray Diffraction, 2.41 A	10	Chimpanzee, Rhesus macaque, Mouse, Rat, Dog, Pig, Cattle, Chicken, Zebrafish
Crenezumab	anti-amyloid antibody	APP	5VZY	X-Ray Diffraction, 2.32 A	12	Chimpanzee, Rhesus macaque, Dog, Pig, Cattle, Chicken
Gantenerumab	anti-amyloid antibody	APP	5CSZ	X-Ray Diffraction, 1.8 A	10	Chimpanzee, Rhesus macaque, Dog, Pig, Cattle, Chicken

Mouse amyloid-beta differs from human at H684R, R676G, Y681F, which changes the binding residues of Aducanumab, Crenezumab, Gantenerumab but not Solanezumab.

5. Interpretation

- Disease mutations concentrate on conserved residues: 111 of 112 pathogenic variants sit at residues identical in all five mammals, a higher share than the background conservation of each protein (Table 5). Conservation supports, but does not by itself establish, that these residues matter.
- The catalytic aspartates of PSEN1 and PSEN2 are identical in every organism tested, including fruit fly and C. elegans.
- For PSEN1 and gamma-secretase drug work, the strong models are Chimpanzee, Rhesus macaque, Mouse, Rat, Dog, Pig, Cattle. They keep the membrane core and all five experimental drug pockets identical to human.
- For APP and amyloid-beta work, the strong models are Chimpanzee, Rhesus macaque, Dog, Pig, Cattle, Chicken. Mouse and rat keep every APP disease residue but differ at three amyloid-beta residues, which changes three of the four antibody-binding sites; amyloid-beta studies in rodents therefore need the human APP sequence (knock-in or transgene).
- Zebrafish and frog are usable with care. Fruit fly and C. elegans are distant models, suited to basic genetics rather than to human-specific mutation or drug studies.

6. Conclusion

We can test PSEN1 and PSEN2 mechanisms and gamma-secretase drugs in mouse and rat, because their presenilins keep the human disease residues, catalytic core and drug-binding residues. For amyloid-beta and anti-amyloid antibodies, non-human primates, pig, dog and cattle carry the human sequence, while rodents need the human APP gene introduced. These are sequence- and structure-level predictions that motivate, but do not replace, experimental validation.

7. Limitations

- Conservation is one line of evidence. Pathogenicity is established clinically, and ClinVar classifications change.
- This analysis is not diagnostic and predicts nothing about any individual.
- Organism structures are AlphaFold predictions; flexible low-confidence regions are excluded from the RMSD.
- Pocket conservation is not docking: identical residues suggest, but do not prove, identical binding.
- Gamma-secretase also contains nicastrin, APH-1 and PEN-2, which were not compared; some drugs contact them.
- The trees use p-distance; substitution-model selection was not performed.

Data, code and AI use

All data come from public databases and are traceable by accession.version and access date. Code, raw downloads and outputs: https://github.com/Kukyos/Nozheimers. Interactive results: https://nozheimers.vercel.app. The scripts, website and this report were produced with the help of an AI assistant (Claude), and every number is computed from the saved outputs. Check your institution's rules on AI use and disclosure.

References

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